

Quantum Leap

Building Real Quantum Programs with Python

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A quantum program in Python

Python	Execution	Result
QuantumCircuit	Local simulator	Counts of bitstrings
The same circuit model	Cloud quantum processor	Counts + job metadata

One shot = one execution and measurement

Qubits and measurement

State: $\alpha|0\rangle + \beta|1\rangle$

$$P(0) = |\alpha|^2 \quad P(1) = |\beta|^2$$

After H on $|0\rangle$: 50% 0, 50% 1

Each measurement gives one bit.

CORRECTION

The old “greater capacity” wording needs this measurement qualification.

The gates used in these notebooks

Gate	Role in our examples
X	Flip 0 and 1
H	Prepare superposition from 0
CX / CCX	Flip a target under one / two controls
$\text{RY}(\theta)$ / CZ	Adjust a trial state / apply a controlled phase

A half-adder

```
qc = QuantumCircuit(4, 2)
qc.x(0)
qc.x(1)
qc.cx(0, 2)
qc.cx(1, 2)
qc.ccx(0, 1, 3)
qc.measure([2, 3], [0, 1])
```

1 + 1 = 10

c1 = carry
c0 = sum

BIT ORDER

Displayed “10” means carry = 1 and sum = 0.

[Documentation](#)

Hello World: three correlated qubits

```
from qiskit import QuantumCircuit
```

```
qc = QuantumCircuit(3)  
qc.h(0)  
qc.cx(0, 1)  
qc.cx(0, 2)  
qc.measure_all()
```

Ideal outcomes

000

111

The circuit prepares an entangled GHZ state.

Local simulation

```
from qiskit.primitives import StatevectorSampler

sampler = StatevectorSampler(seed=42)
job = sampler.run([qc], shots=1024)
counts = job.result()[0].data.meas.get_counts()
print(counts)
```

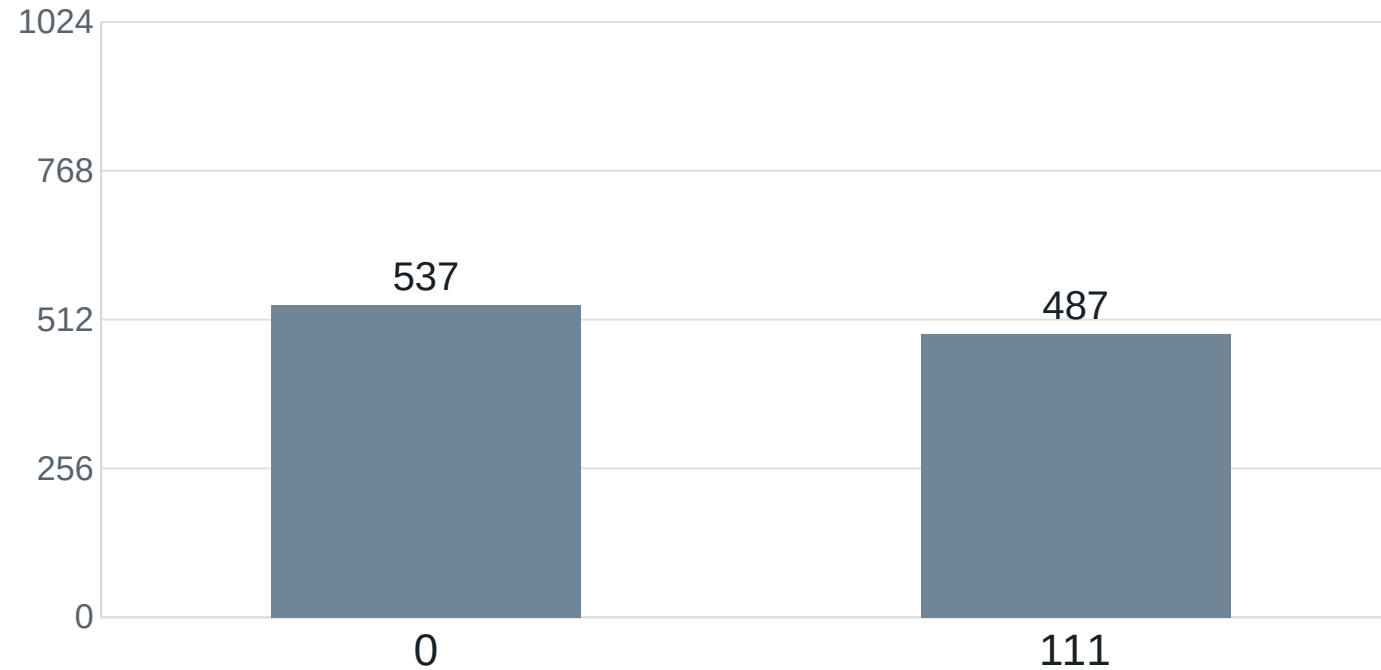
1,024 shots sample the output distribution.

UPDATE

Replaces the old “from qiskit import Aer” execution path.

[Documentation](#)

Reading the counts



$537 + 487$
 $= 1,024$ shots

50/50 describes
probabilities.

SAVED RUN

Counts from Q_Hello_World.ipynb. Rerunning gives a new sample.

A two-bit optimization problem

Maximize $f(a, b) = 3a + 2b - 4ab$

a	b	Displayed q1q0	f(a, b)
0	0	00	0
1	0	01	3
0	1	10	2
1	1	11	1

Best assignment
 $a = 1, b = 0$

BIT ORDER

The rightmost displayed bit is q0, which represents a.

[Documentation](#)

The VQE feedback loop

Step	Where it runs
Choose circuit angles	Classical optimizer
Prepare and measure the trial state	Simulator or quantum processor
Estimate cost and update angles	Classical optimizer

Repeat until the evaluation budget ends.

VQE in the supplied notebook

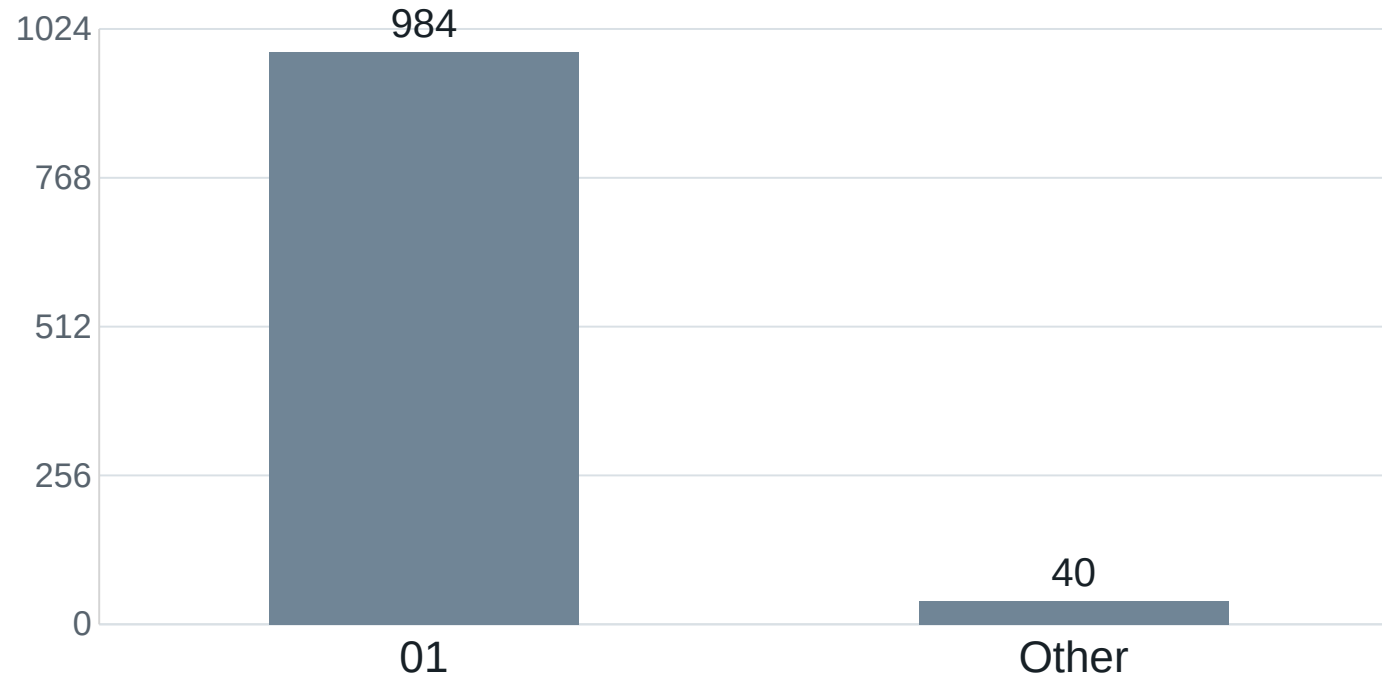
```
algorithm_globals.random_seed = 1234
small_vqe = SamplingVQE(
    sampler=StatevectorSampler(
        default_shots=1024, seed=1234),
    ansatz=small_ansatz,
    optimizer=COBYLA(maxiter=20),
)
small_result = small_vqe.compute_minimum_eigenvalue(
    cost_operator)
```

CURRENT PATTERN

This notebook already uses V2 sampling. Imports and model setup are in VQE_conference.ipynb.

[Documentation](#)

The measured answer



01

$a = 1, b = 0$
 $f(1, 0) = 3$

SAVED RUN

984 of 1,024 measurements give the optimal assignment in this run.

Submitting a cloud job

```
from qiskit_ibm_runtime import (
    QiskitRuntimeService, SamplerV2,
)
from qiskit.transpiler.preset_passmanagers import (
    generate_preset_pass_manager,
)

service = QiskitRuntimeService()
backend = service.least_busy(
    operational=True, simulator=False,
    min_num_qubits=2)
```

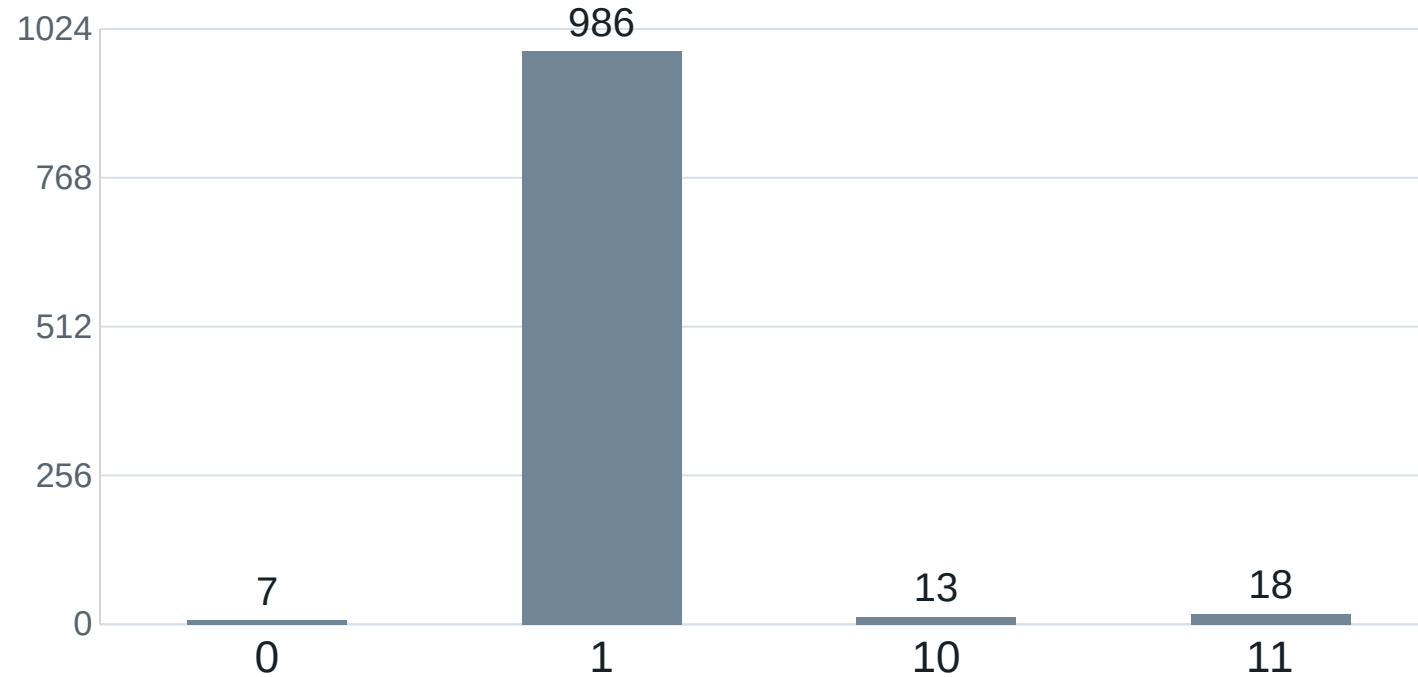
```
# measured_circuit has fixed angles
pm = generate_preset_pass_manager(
    backend=backend,
    optimization_level=3,
)
isa = pm.run(measured_circuit)
job = SamplerV2(mode=backend).run(
    [isa], shots=1024,
)
print(job.job_id())
```

CHECK BEFORE TALK

Account, instance access, backend availability and queue. Use saved results on stage.

[Documentation](#)

The saved hardware result



01: 986 / 1,024

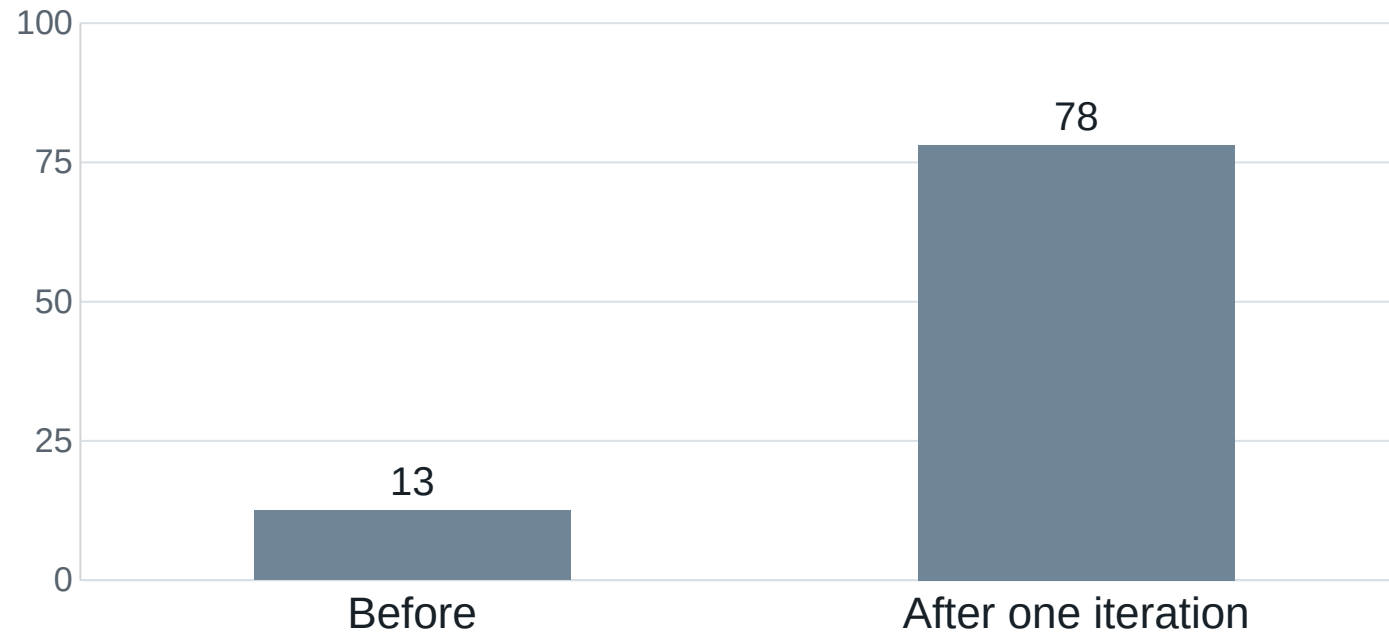
Local simulator:
984 / 1,024

Angles trained locally

SAVED RUN

ibm_marrakesh. Fixed trained circuit, one sampling job. Recheck job metadata.

Grover: increasing a target probability



Target: 010
Eight possible strings

Ideal probabilities (%)
calculated from this circuit

UPDATE

GroverOperator is deprecated. Use grover_operator. Oracle queries are the complexity unit.

[Documentation](#)

Route optimization with QAOA

Moscow / Baku / Doha / Florianópolis

Illustrative cost: $4 + 2 + 5 = 11$

Encoding	Saved sample distribution
8 edges = 8 binary variables = 8 qubits	1.9% feasible outcomes

CORRECTION

TSP.ipynb solves a constrained shortest path. Its prose still describes an older graph.

[Documentation](#)

What these examples establish

Example	What we can observe
Hello World	Correlated measurement outcomes
VQE	A dominant optimum in one small run
Cloud execution	Samples from a real device
QAOA route	A feasible answer amid many invalid samples

These small runs do not establish quantum advantage.

Code and questions

github.com/beilak/qdev

[Notebook repository](#)

Questions

CHECK BEFORE TALK

Confirm that the public repository contains the exact notebook versions you will present.

Backup: the larger VQE example

5 model variables / 14 encoded qubits

Saved assignment: (1, 1, 0, 10, 5)

Objective: 47

Best feasible sampled candidate can differ from the most frequent bitstring.

INTERPRETATION

A printed optimum alone does not show that the circuit concentrates on that answer.

[Documentation](#)

Backup: API changes to review

In the old material	Pattern used in this draft
<code>from qiskit import Aer</code>	StatevectorSampler for ideal sampling
<code>IBMPProvider + backend.run</code>	QiskitRuntimeService + SamplerV2
<code>GroverOperator(...)</code>	<code>grover_operator(...)</code>
<code>qiskit['optimize']</code>	Use explicit supported packages

UPDATE

Check the rehearsed environment after any package update.

[Documentation](#)

Backup: content that needs review

Item	Action before presenting
433 qubits / Osprey	Historical: 2022. Correct the ItFest date.
“Greater capacity”	Explain amplitudes and one-bit measurement.
TSP text / seven qubits	Use shortest path / eight qubits.
QML labels, biography, contacts	Review relevance and current details.

CHECK BEFORE TALK

Hardware counts and old dates describe historical material.

[Documentation](#)

Reference links

[Qiskit installation and supported environment](#)

[Local sampling and measurement results](#)

[IBM account setup](#)

[SamplerV2 cloud execution](#)

[SamplingVQE](#)

[QUBO conversion and sample interpretation](#)

[Grover operator and query model](#)

[Qiskit bit ordering](#)

Checked 27 September 2026